21 December 1972
SUPERSEDING
MIL-H-22832A
22 September 1964

MILITARY SPECIFICATION

HEAT INTERCHANGERS, REFRIGERANT 12

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers commercial, liquid to suction gas heat interchangers for use with refrigerant 12 (dichlorodifluoromethane).

1.2 Classification. The heat interchangers shall be of the following sizes as specified (see 6.2):

Size	Capacity (minimum) Tons of Refrigeration (T.R)
1	1/12
2	1/6
3	1/4
4	1/3
5	1/2
6	3/4
7	1
8	2

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

WW-T-799 - Tube, Copper, Seamless, (For Use With Solder-Joint or Flared-Tube Fittings).

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PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Set-Up, Paperboard.

MILITARY

MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MS 17241 - Heat Interchanger, Refrigerant Type 12.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

★ 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Underwriters' Laboratories, Inc. (UL)

No. 207 (c) - Refrigerant-Containing Components

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611; Walt Whitman Road, Melville, L.I. New York 11749 or 1655 Scott Blvd., Santa Clara, California 95050.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

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Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), the supplier shall furnish a sample for first article inspection and approval (see 4.2, and 6.3).

* 3.2 Standard product. The heat interchangers delivered under this specification shall be the manufacturer's standard commercial product, except for any changes necessary to meet specification requirements. Component parts of the heat interchanger need not be the product of the same manufacturer.

3.3 Codes and standard. Heat interchangers shall conform to the applicable requirements of UL Standard No. 207 (c).

* 3.3.1 Certification. Prior to approval of the first article if one is submitted, or prior to commencing production, the supplier shall submit to the contracting officer of his authorized representative satisfactory evidence that the heat interchangers he proposes to furnish under this specification meet the applicable requirements of UL Standard No. 207 (c). Acceptable evidence of meeting the requirements of this standard shall be the UL label, or a UL listing mark certified test report from a recognized independent testing laboratory, indicating that the heat interchangers offered conform to the requirements of UL Standard No. 207 (c).

* 3.4 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer for heat interchangers provided the completed item complies with all provisions of this specification.

3.4.1 Copper. Copper tubing shall conform to type K or L of WW-T-799.

3.5 Design and construction. The heat interchangers shall be designed and constructed in accordance with MS 17241 and as specified herein. The gas circuit shall be arranged to eliminate entrapment of oil. The liquid circuit and gas nipples shall be fabricated of the copper tube specified in 3.4.1.

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3.5.1 Capacity. The heat interchanger shall have the capacities indicated in 1.2 under the conditions specified on MS 17241 (see 4.4.1).

* 3.6 Identification marking. The manufacturers' name or identifiable trade name, shall be clearly stamped, embossed, or etched on a securely attached corrosion-resistant metal plate. In lieu of the metal identification plate, a pressure-sensitive label or waterproofed decalcomania may be used. In addition, the heat interchanger shall be marked as required by UL Standard No. 207 (c).

3.7 Finish. The heat interchanger shall have the manufacturer's standard commercial finish. The finished heat interchanger shall not have any sharp burrs, slivers, or splinters.

* 3.8 Workmanship. The finished product shall not be fractured, split, punctured, dented, or malformed. Soldering shall be complete, clean, and adherent and shall not have any pinholes.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article. When a first article is required, it shall be examined for the defects listed in table I and for dimensions specified herein, and tested as specified in 4.4. Failure of any test shall be cause for rejection of the first article. The presence of any defects listed in table I, or any dimension not within specified requirements shall be cause for rejection of the first article.

* 4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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4.3.2 In-process inspection. Examination shall be made that the gas circuit is arranged to eliminate entrapment of oil as specified in 3.5. Whenever nonconformance is noted, correction shall be made to the affected items and lot in process.

* 4.3.3 End item inspection. The lot shall consist of all the heat interchangers of one size offered for inspection at one time. The sample unit for the inspection shall be one heat interchanger.

* 4.3.3.1 Visual examination. Examination shall be made of the heat interchanger for defects set forth in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Standard product	Item not in accordance with manufacturer's standard product except for changes necessary to comply with specification requirements	X	
Finish	Finish not manufacturer's standard type		X
	Sharp burrs, slivers, splinters, or rough edges	X	
Construction, workmanship (applicable to all components and assemblies)	Fractured, split, punctured, dented, malformed	X	
	Part missing	X	
Soldering	Soldering not clean (flux or flux residue not removed): pin holes in solder, not complete, not adherent	X	
Marking	Identification marking missing, not legible, not specified type, not affixed in the prescribed manner		X

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* 4.3.3.2 Dimensional examination. Examination shall be made of the end item to determine compliance with dimensional requirements. Any dimension that is not within the specified requirements shall be classified a defect. The inspection level shall be S-4 and the AQL shall be 2.5 expressed in terms of defects per hundred units.

* 4.3.3.3 End item testing. When a first article is not required, one unit from the initial production lot shall be tested as specified in 4.4.1. Any nonconformance with specified requirements shall be cause for rejection of the lot.

* 4.3.4 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be as indicated in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

TABLE II. Preparation for delivery defects

Examination	Defect
Markings	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components such as incomplete closure of case liners.
Contents	Content per container is more or less than required.

* 4.3.4.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-636 or PPP-B-621, examination for defects in closure, waterproofing, and strapping shall be in accordance with the applicable container specification.

* 4.4 Tests.

4.4.1 Capacity test. A capacity test shall be performed to determine compliance with 3.5.1. The heat interchanger shall be connected to a refrigeration system having the T. R. capacity of the heat interchanger to be tested.

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When the system is placed into operation the heat interchanger shall be observed to insure that it has the capacity and capability of raising the suction gas temperature from minus 5° Fahrenheit (F), at a pressure corresponding to minus 10° F. saturated, to plus 30° F. superheated leaving temperature, with the entering liquid temperature not higher than 100° F. During the test, the liquid circuit pressure drop shall be negligible. The gas circuit pressure drop shall not exceed 1/4 pound per square inch (psi) at minus 10° F. evaporating temperature and 1/2 psi at plus 25° F. evaporating temperature.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each heat interchanger shall be packaged in a snug-fitting fiberboard, folding paperboard, or set-up paperboard box conforming to style RSC (end opening) or FPF, type CF or SF, class domestic, variety optional of PPP-B-636; style III, type B, class a or style IV, type A or B, class a of PPP-B-566; or type I, grade A, class A, style 2 of PPP-B-676. Closure shall be in accordance with the appendix of the applicable box specification.

5.1.2 Level C. Heat interchangers shall be packaged in accordance with industry's practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A. Heat interchangers, packaged as specified in 5.1, shall be packed in the quantity as specified in table III in a snug-fitting fiberboard or nailed wood shipping container conforming to style RSC, V3c or V3s of PPP-B-636, except that the weight limitation shall not be exceeded, or class 2, style 2 or 4 of PPP-B-621, respectively. Each fiberboard shipping container shall be closed, waterproofed by means of tape, and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification. Each nailed wood shipping container shall be provided with a type I or II, grade C case liner conforming to MIL-L-10547, and closed and strapped in accordance with the appendix of the container specification.

TABLE III. Quantity of heat interchangers per container

Size	Quantity per shipping container
1	72
2	60
3	24
4	24
5	24
6	12
7	6
8	6

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* 5.2.2 Level B. Heat interchangers, packaged as specified in 5.1, shall be packed in the quantity as specified in table III in a snug-fitting fiberboard shipping container conforming to style RSC, type CF or SF, class domestic, variety optional of PPP-B-636. Closure shall be in accordance with appendix of the container specification.

5.2.3 Level C. Heat interchangers, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The heat interchangers covered by this specification are intended for use on portable or stationary mechanical refrigeration systems.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) When a first article is required (see 3.1, 4.2, and 6.3).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Armed Services Procurement Regulations ASPR 7-104.55. The first article should be a preproduction sample, initial production item, or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations.

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